

Conversion of Units within the Metric System

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The **International System of Units (SI)** is the standard system of measurement used worldwide.

Converting Larger Units

To convert from a **larger unit** to a **smaller unit**, multiply by the appropriate power of 10.
Move the decimal point to the right.

$$4.8 \text{ km} = 4800 \text{ m}$$

Converting Smaller Units

To convert from a **smaller unit** to a **larger unit**, divide by the appropriate power of 10.
Move the decimal point to the left.

$$3600 \text{ cm} = 36 \text{ m}$$

Metric Prefixes

kilo (k) = 1000

hecto (h) = 100

deka (da) = 10

base unit = meter (m), gram (g), liter (L)

Metric Prefixes

$$\begin{aligned}\text{deci (d)} &= \frac{1}{10} \\ \text{centi (c)} &= \frac{1}{100} \\ \text{milli (m)} &= \frac{1}{1000}\end{aligned}$$

Conversion Ladder

A useful conversion ladder is:

k $\rightarrow$ **h** $\rightarrow$ **da** $\rightarrow$ **Unit** $\rightarrow$ **d** $\rightarrow$ **c** $\rightarrow$ **m**

Each step is equivalent to multiplying or dividing by 10.

Examples

Example 1

Convert 4.8 km to m.

$$4.8 \text{ km} \times 1000 = 4800 \text{ m}$$

Therefore, $4.8 \text{ km} = 4800 \text{ m}$.

Example 2

Convert 3600 cm to m.

$$3600 \text{ cm} \div 100 = 36 \text{ m}$$

Therefore, $3600 \text{ cm} = 36 \text{ m}$.

Example 3

Convert 5.2 kg to g.

$$5.2 \text{ kg} \times 1000 = 5200 \text{ g}$$

Therefore, $5.2 \text{ kg} = 5200 \text{ g}$.

Practice Exercises

Practice 1

Convert 4.8 km to m.

$$4.8 \text{ km} \rightarrow \text{m}$$

Practice 2

Convert 3600 cm to m.

$$3600 \text{ cm} \rightarrow \text{m}$$

Practice 3

Convert 5.2 kg to g.

$$5.2 \text{ kg} \rightarrow \text{g}$$

Practice 4

Convert 8450 mL to L.

$$8450 \text{ mL} \rightarrow \text{L}$$

Practice 5

Convert 7.25 m to cm.

$$7.25 \text{ m} \rightarrow \text{cm}$$

Practice 6

Convert 9800 mg to g.

$$9800 \text{ mg} \rightarrow \text{g}$$

Practice 7

Convert 3.4 L to mL.

$$3.4 \text{ L} \rightarrow \text{mL}$$

Practice 8

Convert 5600 g to kg.

$$5600 \text{ g} \rightarrow \text{kg}$$

Practice 9

Convert 0.85 m to mm.

$$0.85 \text{ m} \rightarrow \text{mm}$$

Practice 10

Convert 125000 mm to m.

$$125000 \text{ mm} \rightarrow \text{m}$$

Activities

Activity 1

Convert 6.3 km to m.

$$6.3 \text{ km} \rightarrow \text{m}$$

Activity 2

Convert 4250 cm to m.

$$4250 \text{ cm} \rightarrow \text{m}$$

Activity 3

Convert 8.4 kg to g.

$$8.4 \text{ kg} \rightarrow \text{g}$$

Activity 4

Convert 9600 mL to L.

$$9600 \text{ mL} \rightarrow \text{L}$$

Activity 5

Convert 9.15 m to cm.

$$9.15 \text{ m} \rightarrow \text{cm}$$

Activity 6

Convert 12600 mg to g.

$$12600 \text{ mg} \rightarrow \text{g}$$

Activity 7

Convert 2.75 L to mL.

$$2.75 \text{ L} \rightarrow \text{mL}$$

Activity 8

Convert 8700 g to kg.

$$8700 \text{ g} \rightarrow \text{kg}$$

Activity 9

Convert 1.2 m to mm.

$$1.2 \text{ m} \rightarrow \text{mm}$$

Activity 10

Convert 98000 mm to m.

$$98000 \text{ mm} \rightarrow \text{m}$$

Thank you!